A note on the Sherrington-Kirkpatrick model

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This is an informal note on some proofs of β-thresholds for a Poincaré inequality for the Sherrington Kirkpatrick model (Ising with GOE interaction matrix). Section 1 starts by recalling the proof in [3] of $\beta < 1/4$ (hypercube). Section 2 presents a cute AI-generated proof of $\beta < 1$ in the spherical SK setting. Section 3 adapts the idea (second-order Lyapunov argument; the explicit Lyapunov function was found by AI) from the sphere back to the hypercube to obtain a $\beta < 0.37$ threshold, which was quickly superseded by proofs of $\beta < 1/2$ obtained in [8, 18] around August 2026, and then later by the (fully resolved, it appears) $\beta < 1$ proof by OpenAI [15] in October 2026.

As this document is not published (and will not be submitted for publication), it should be considered more a sketch of interesting ideas rather than a credible source. I am sure there are typos or some minor errors here and there, but I believe the overall arguments ($\beta < 1$ for sphere, $\beta < 0.37$ for hypercube) to be correct, and illuminating (in particular, they make more use of GOE in the form of local laws and rigidity of eigenvalues, than the previous bounds in [2,3], and even in the superseding $\beta < 1/2$ bound [18] who use only the spectral width of the GOE). The exposition is also rather light (or perhaps nonexistent) and I have not thought about this problem since August 2026. But, if you spot any mistakes or there is any demand for an exposition, you may email me and I will be happy to fix it.

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1. The $\beta < \frac{1}{4}$ argument

sec:beta-quarter

Let $\Lambda = [\![N]\!]$ be a finite set of N sites, and $\boldsymbol{H} = (\boldsymbol{H}_{xy})_{x,y\in\Lambda}$ be a symmetric positive definite matrix. Recall the $O(n)$ model,

$$\nu_{\mathrm{SK}}(d\sigma) = \frac{1}{Z_{\mathrm{SK}}} e^{-\frac{1}{2}\langle \sigma, \boldsymbol{H}\sigma\rangle} \prod_{x\in\Lambda} \rho_n(d\sigma_x), \tag{1.1}$$

where $\rho_n = \rho_{n,r}$ is the surface measure on $r\mathbb{S}^{n-1} \subset \mathbb{R}^n$. Note that the measure ν_{SK} is invariant under shifts $\boldsymbol{H} \mapsto \boldsymbol{H} + c\,\mathrm{id}$. For $F \in C^1(\mathbb{S}^{n-1})$ we let ∇F be the covariant derivative on $\mathbb{S}^{n-1}$ if $n > 1$ and $|\nabla F|^2 = |F(\sigma) - F(-\sigma)|^2$ if $n = 1$. The Sherrington-Kirkpatrick (SK) model is the case $n = 1$ and we take M to be a GOE matrix whose entries have variance $1/N$ above the diagonal. For the rest of this note we focus only on this case, though we remark that the results apply more generally as in [3].

A related model is the spherical Sherrington-Kirkpatrick (SSK) model, in which we replace the reference measure on the hypercube $\{\pm 1\}^N$ with the sphere $\sqrt{N}\mathbb{S}^{N-1}$,

$$\nu_{\mathrm{SSK}}(d\sigma) = \frac{1}{Z_{\mathrm{SSK}}} e^{-\frac{1}{2}\langle \sigma, \boldsymbol{H}\sigma\rangle} \rho_{N,\sqrt{N}}(d\sigma). \tag{1.2}$$

Theorem 1.1. *Assume $\boldsymbol{H}$ is positive definite and $\|\boldsymbol{H}\|_{op} < 1$. Then $\nu = \nu_{SK}$ satisfies an LSI,*

$$\mathrm{Ent}_\nu(F^2) \le \frac{2}{\gamma}\left(1 + \frac{\|\boldsymbol{H}\|_{op}}{1-\|\boldsymbol{H}\|_{op}}\right)\mathbb{E}_\nu\left[\sum_{x\in\Lambda}|\nabla_{\sigma_x}F|^2\right]. \tag{1.3}$$

Here, the constant $\gamma = 4$ is the worst case constant for which the one-dimensional LSI,

$$\mathrm{Ent}_{\rho_1^h}(F^2) \le \frac{2}{\gamma}\mathbb{E}_{\rho_1^h}\left[|F(\sigma) - F(-\sigma)|^2\right], \tag{1.4}$$

holds for all single spin measures $\rho_1^h = \mathrm{Ber}(p)$ on $\{\pm 1\}$.

Proof. We run a Polchinski dynamics with $\nu_0 = \nu$ and a covariance decomposition $\boldsymbol{H}^{-1} = C_\infty = \int_0^s \mathrm{id}\, du + C_{s,\infty}$. Note that s must satisfy

$$C_s = s\,\mathrm{id} \preceq \boldsymbol{H}^{-1}, \tag{1.5}$$ e:Cs-H-req

i.e. $\boldsymbol{H} \preceq s^{-1}\,\mathrm{id}$. The starting potential V_0 is nothing but the uniform (product) measure on the hypercube. In particular, for all times $0 \le u \le s$ the renormalized potential V_s also splits into a sum of potentials $\sum_{x\in\Lambda} U_s(\sigma_x)$.

Then, we decompose the entropy,

$$\mathrm{Ent}_{\nu_0}\left(F^2\right) = \mathbb{E}_{\nu_s}\left[\mathrm{Ent}_{\mu_s^\varphi}\left(F^2\right)\right] + \mathrm{Ent}_{\nu_s}\left(\mathbb{E}_{\mu_s^\varphi}\left[F^2\right]\right). \tag{1.6}$$ e:ent-decomp-1

Since $C_s = s\,\mathrm{id}$ is diagonal, the fluctuation measure $d\mu_s^\varphi(\sigma) = \exp(-\frac{1}{2}|\sigma-\varphi|^2_{C_s^{-1}})\prod_{x\in\Lambda} d\rho_1(\sigma_x)$ is also a product measure on the hypercube $\{\pm 1\}^N$. In particular, by tensorization we may bound the first term in (1.6) by

$$\mathbb{E}_{\nu_s}\left[\mathrm{Ent}_{\mu_s^\varphi}\left(F^2\right)\right] \le \frac{2}{\gamma}\mathbb{E}_{\nu_0}\left[\sum_{x\in\Lambda}|\nabla_{\sigma_x}F|^2\right]. \tag{1.7}$$ e:ent-a-est

For the second term, we start by writing (denoting ∂_φ the Euclidean gradient in $\mathbb{R}^N$)

$$\partial_\varphi\sqrt{\mathbb{E}_{\mu_s^\varphi}[F^2]} = \frac{\partial_\varphi\mathbb{E}_{\mu_s^\varphi}[F^2]}{2\sqrt{\mathbb{E}_{\mu_s^\varphi}[F^2]}} = \frac{1}{2s}\frac{\mathrm{Cov}_{\mu_s^\varphi}\left(F^2,\sigma\right)}{\sqrt{\mathbb{E}_{\mu_s^\varphi}[F^2]}}. \tag{1.8}$$ e:ent-a-est

In particular,

$$\left|\partial_\varphi\sqrt{\mathbb{E}_{\mu_s^\varphi}[F^2]}\right|^2 = \frac{1}{4s^2\mathbb{E}_{\mu_s^\varphi}[F^2]}\sum_{y\in\Lambda}\mathrm{Cov}_{\mu_s^\varphi}\left(F^2,\sigma_y\right)^2.$$

Denoting $\overset{1}{\sigma}$ and $\overset{2}{\sigma}$ two i.i.d. "replicas" distributed according to μ_s^φ, we apply Cauchy-Schwarz and the fact that $\sigma_y^2 \le 1$ to see that

$$\begin{aligned}
\mathrm{Cov}_{\mu_s^\varphi}\left(F^2,\sigma_y\right) &= \mathbb{E}_{(\mu_s^\varphi)^{\otimes 2}}\left[F(\overset{1}{\sigma})^2(\overset{1}{\sigma}_y - \overset{2}{\sigma}_y)\right] \\
&= \mathbb{E}_{(\mu_s^\varphi)^{\otimes 2}}\left[\left(F(\overset{1}{\sigma})^2 - F(\overset{2}{\sigma})^2\right)\overset{1}{\sigma}_y\right] \\
&\le \mathbb{E}_{(\mu_s^\varphi)^{\otimes 2}}\left[\left(F(\overset{1}{\sigma}) - F(\overset{2}{\sigma})\right)^2\right]^{1/2}\mathbb{E}_{(\mu_s^\varphi)^{\otimes 2}}\left[\left(F(\overset{1}{\sigma}) + F(\overset{2}{\sigma})\right)^2\right]^{1/2} \\
&\le \mathrm{Var}_{\mu_s^\varphi}(F)^{1/2}\left(4\mathbb{E}_{\mu_s^\varphi}\left[F(\sigma)^2\right]\right)^{1/2}.
\end{aligned}$$

By the spectral gap inequality implied by the LSI for μ_s^φ (with constant γ, by tensorization as above), we have

$$\mathrm{Cov}_{\mu_s^\varphi}\left(F^2,\sigma_y\right)^2 \le \frac{4}{\gamma}\mathbb{E}_{\mu_s^\varphi}\left[\sum_{x\in\Lambda}|\nabla_{\sigma_x}F|^2\right]\mathbb{E}_{\mu_s^\varphi}\left[F^2\right], \tag{1.9}$$

and hence

$$\left|\partial_\varphi \sqrt{\mathbb{E}_{\mu_s^\varphi}[F^2]}\right|^2 \leq \frac{|\Lambda|}{s^2\gamma}\mathbb{E}_{\mu_s^\varphi}\left[\sum_{x\in\Lambda}|\nabla_{\sigma_x}F|^2\right]. \tag{1.10}$$

But this is a factor of $|\Lambda| = N$ too bad; in [3] what they do to remedy this is to instead replicate only the spin σ_y at site y while keeping the others $\sigma_{\Lambda\setminus\{y\}}$ fixed, i.e. we consider two replicas $\overset{1}{\sigma}$ and $\overset{2}{\sigma}$ that are only independent in the y-th site

$$\overset{1}{\sigma} = \left[\sigma_{\Lambda\setminus y}, \overset{1}{\sigma}_y\right],$$
$$\overset{2}{\sigma} = \left[\sigma_{\Lambda\setminus y}, \overset{2}{\sigma}_y\right].$$

Then by independence of the sites (the fluctuation measure is constructed as such)

$$\mathrm{Cov}_{\mu_s^\varphi}(F^2,\sigma_y) = \mathrm{Cov}_{\overset{2}{\sigma}}(F^2,\sigma_y) = \mathbb{E}_{\overset{1}{\sigma}}\left[\mathrm{Cov}_{\overset{2}{\sigma}_y}\left(F^2,\overset{2}{\sigma}_y\right)\right]. \tag{1.11}$$

Now, with $\sigma_{\Lambda\setminus\{y\}}$ fixed and omitting it in the notation, write

$$\begin{aligned}\mathrm{Cov}_{\overset{2}{\sigma}_y}\left(F^2,\overset{2}{\sigma}_y\right) &= \mathbb{E}\left[F(\overset{2}{\sigma}_y)^2\left(\overset{2}{\sigma}_y - \overset{1}{\sigma}_y\right)\right]\\ &= \mathbb{E}\left[\left(F(\overset{2}{\sigma}_y)^2 - F(\overset{1}{\sigma}_y)^2\right)\overset{1}{\sigma}_y\right]\\ &\leq \mathrm{Var}_{\overset{1}{\sigma}_y}(F)^{1/2}\left(4\mathbb{E}_{\overset{1}{\sigma}_y}\left[F^2\right]\right)^{1/2}.\end{aligned}$$

Applying Cauchy-Schwarz, this now yields the better bound (localizing the gradient to only ∇_{σ_y})

$$\mathrm{Cov}_{\mu_s^\varphi}\left(F^2,\sigma_y\right)^2 \leq \frac{4}{\gamma}\mathbb{E}_{\mu_s^\varphi}\left[|\nabla_{\sigma_y}F|^2\right]\mathbb{E}_{\mu_s^\varphi}\left[F^2\right], \tag{1.12}$$

and therefore

$$\left|\partial_\varphi \sqrt{\mathbb{E}_{\mu_s^\varphi}[F^2]}\right|^2 \leq \frac{1}{s^2\gamma}\mathbb{E}_{\mu_s^\varphi}\left[\sum_{y\in\Lambda}|\nabla_{\sigma_y}F|^2\right] \tag{1.13}$$

Now, the measure ν_s is of the form

$$d\nu_s(\varphi) \propto \exp\left(-\frac{1}{2}|\varphi|^2_{C_{s,\infty}^{-1}} - V_s(\varphi)\right)d\varphi, \tag{1.14}$$

where the smoothed potential V_s has a diagonal Hessian which satisfies (using the fact that $\mathrm{Var}_{\mu_s^\varphi}(\sigma_x) \leq 1$)

$$\mathrm{Hess}\, V_s(\varphi) = C_s^{-1} - C_s^{-1}\,\mathrm{Cov}(\mu_s^\varphi)C_s^{-1} \succeq \frac{1}{s} - \frac{1}{s^2}. \tag{1.15}$$

Hence, taking $s^{-1} < 1$ so that V_s is convex, the requirement (1.5) forces $\|H\|_{\mathrm{op}} < 1$. The Bakry-Émery curvature then tells us that $C_{\mathrm{LSI}}(\nu_s) \leq 2/(\|H\|_{\mathrm{op}} - \|H\|^2_{\mathrm{op}})$, so that

$$\mathrm{Ent}_{\nu_s}\left(\mathbb{E}_{\mu_s^\varphi}\left[F^2\right]\right) \leq \frac{2}{\|H\|_{\mathrm{op}}\left(1-\|H\|_{\mathrm{op}}\right)}\cdot\frac{\|H\|^2_{\mathrm{op}}}{\gamma}\mathbb{E}_{\nu_0}\left[\sum_{x\in\Lambda}|\nabla_{\sigma_x}F|^2\right], \tag{1.16}$$ e:ent-b-est

so altogether,

$$\mathrm{Ent}_{\nu_0}\left(F^2\right) \leq \frac{2}{\gamma}\left(1 + \frac{\|H\|_{\mathrm{op}}}{1-\|H\|_{\mathrm{op}}}\right)\mathbb{E}_{\nu_0}\left[\sum_{x\in\Lambda}|\nabla_{\sigma_x}F|^2\right], \tag{1.17}$$ e:ent-b-est

as desired. □

1.1. An improvement by a Lyapunov argument. We need a better estimate of $C_{\mathrm{PI}}(\nu_s)$ that uses more than just the bare Hessian estimate,

$$\nabla^2 U_s(\sigma) = C_{s,\infty}^{-1} + \frac{1}{s}\,\mathrm{id} - \frac{1}{s^2} A_s^\sigma \succeq \left(\frac{1}{s} - \frac{1}{s^2}\right)\mathrm{id}\,. \tag{1.18}$$

A first step towards this direction is to improve this Hessian estimate *in expectation*. In particular, we are actually afforded to estimate the form,

$$\mathbb{E}_{\nu_s}\left[\Gamma_{2,s}(f)\right] = \mathbb{E}_{\nu_s}\left[|\nabla^2 f|^2 + \left(\nabla f, \left[\nabla^2 U_s\right]\nabla f\right)\right]. \tag{1.19}$$

Let $\nu_{s,i}^J = \nu_{s,i}^{\{\sigma_j\}_{j\neq i}}$ denote the one-dimensional conditional measures,

$$\nu_{s,i}^{\{\sigma_j\}_{j\neq i}}(d\sigma_i) \propto \exp\left(-v_s(\sigma_i) - \frac{1}{2}B_{ii}\sigma_i^2 - \sum_{j\neq i} B_{ij}\sigma_i\sigma_j\right) d\sigma_i, \tag{1.20}$$

where we denoted $B = B_s = C_{s,\infty}^{-1}$, and $V_s(\sigma) = \sum_{i=1}^N v_s(\sigma_i)$ the one-dimensional components of the renormalized potential. Denoting $\nu_{s,J}$ the $\sigma_J = \{\sigma_j\}_{j\neq i}$ marginals of ν_s, note that the (one-dimensional) spectral gap for $\nu_{s,i}^J$ implies the $\Gamma_{2,i}^J > \Gamma_{1,i}^J$ estimate:

$$\mathbb{E}_{\nu_{s,i}^J}\left[(f'')^2 + \left[U_{s,i}^J\right]''(f')^2\right] \geq C_{\mathrm{PI}}(\nu_{s,i}^J)\mathbb{E}_{\nu_{s,i}^J}\left[(f')^2\right]. \tag{1.21}$$

Therefore we can discard the positive terms B_s and off-diagonals $(\partial_i\partial_j f)^2\mathbb{1}_{j\neq i}$ to estimate

$$\begin{aligned}
\mathbb{E}_{\nu_s}\left[\Gamma_2(f)\right] &\geq \sum_{i=1}^N \mathbb{E}_{\nu_s}\left[(\partial_{ii}f)^2 + v_s''(\sigma_i)(\partial_i f)^2\right] \\
&= \sum_{i=1}^N \mathbb{E}_{\nu_{s,J}}\left[\mathbb{E}_{\nu_{s,i}^J}\left[\Gamma_{2,i}^J(f^J) - B_{ii}\left(\partial_{\sigma_i} f(\sigma_i,\sigma_J)\right)^2\right]\right] \\
&\geq \sum_{i=1}^N \mathbb{E}_{\nu_{s,J}}\left[\left\{C_{\mathrm{PI}}(\nu_{s,i}^J) - B_{ii}\right\}\mathbb{E}_{\nu_{s,i}^J}\left[\Gamma_{1,i}^J(f^J)\right]\right] \\
&\geq \mathbb{E}_{\nu_s}\left[\Gamma_{1,s}(f)\right]\cdot \min_{1\leq i\leq N}\inf_{h\in\mathbb{R}}\left\{C_{\mathrm{PI}}\left(\nu_{s,i}^h\right) - B_{ii}\right\}
\end{aligned}$$

where

$$\nu_{s,i}^h(d\sigma_i) \propto \exp\left(-v_s(\sigma_i) - \frac{1}{2}B_{ii}\sigma_i^2 - h\sigma_i\right) d\sigma_i. \tag{1.22}$$

Now, apparently a 1-d Lyapunov argument analysis on $\nu_{s,i}^h$, optimized together with B_{ii}, yields $\beta \lesssim 0.39$.

2. Spherical Sherrington-Kirkpatrick model

sec:SSK

In this section, we prove a Poincaré inequality for spherical Sherrington-Kirkpatrick (SSK) models,

$$\nu_H^\varphi(d\sigma) = \nu_{\beta,H}^\varphi(d\sigma) = \frac{1}{Z_{\beta,H}^\varphi} \exp\left(-\beta U_H^\varphi(\sigma)\right) dS(\sigma), \qquad U_H^\varphi(\sigma) := \frac{1}{2}\left(\sigma-\varphi, H(\sigma-\varphi)\right), \tag{2.1}$$

d:SSK

where $S(d\sigma)$ is the surface measure on $\mathbb{S}^{N-1}_{\sqrt{N}}$, and $\varphi \in \mathbb{R}^N$. The case $\varphi = 0$ corresponds to the standard SSK model without external field. Note that the measure ν_H^φ has the equivalent form,

$$\nu_H^\varphi(d\sigma) \propto \exp\left(-\frac{\beta}{2}(\sigma, \widetilde{H}\sigma) + \beta(H\varphi, \sigma)\right) dS(\sigma), \tag{2.2}$$

d:SSK-ext-h

where $\widetilde{H} = H + c\,\mathrm{id}$ is the equivalence class of all scalar id shifts of the matrix H. To shorten the notation, we use the notation $S = \mathbb{S}^{N-1}_{\sqrt{N}}$, and more generally $S_r = \mathbb{S}^{N-1}_r$. We denote by π_σ the $N \times N$ matrix projecting to the tangent space T_σ, i.e. if $|\sigma| = r$ then

$$\pi_\sigma = \mathrm{id}_{N\times N} - \frac{1}{N}\sigma \otimes \sigma. \tag{2.3}$$

d:SSK-ext-h

To the equilibrium measure ν_H^φ we have the reversible diffusion operator,

$$L_H^\varphi = \Delta_S - \beta \nabla_S U_H^\varphi \cdot \nabla_S \tag{2.4}$$

$$= \Delta_S - \beta\left(\pi_\sigma\left(\widetilde{H}\sigma - H\varphi\right)\right)\cdot \nabla_S, \tag{2.5}$$

the associated $\Gamma_1^{H,\varphi}$ and $\Gamma_2^{H,\varphi}$ operators,

$$\Gamma_1^{H,\varphi}(f) = |\nabla_S f|^2, \tag{2.6}$$

$$\Gamma_2^{H,\varphi}(f) = |\nabla_S^2 f|^2 + \left(\nabla_S f, \left[\mathrm{Ric}_S + \beta\nabla_S^2 U_H^\varphi\right]\nabla_S f\right) \tag{2.7}$$

$$= |\nabla_S^2 f|^2 + \left\{\frac{N-2}{N} - \frac{\beta}{N}(\sigma, \widetilde{H}\sigma) + \frac{\beta}{N}(\sigma, H\varphi)\right\}|\nabla_S f|^2 + \beta\left(\nabla_S f, \left[\widetilde{H}\right]\nabla_S f\right). \tag{2.8}$$

Note that for $\varphi = 0$ we may use the crude bounds $(\sigma, H\sigma) \le N\lambda_{\max}(H)$ and $(\nabla_S f, H\nabla_S f) \ge |\nabla_S f|^2\lambda_{\min}(H)$ which for $H \sim$ GOE already yields the $\beta < 1/4$ threshold for the Poincaré (or log-Sobolev) inequality. To do better we must handle the scalar curvature term,

$$\left\{1 - \frac{\beta}{N}(\sigma, \widetilde{H}\sigma) + \frac{\beta}{N}(\sigma, H\varphi)\right\}\Gamma_1^{H,\varphi}(f), \tag{2.9}$$

e:sphere-ibp-term

through an integration by parts.

l:LS-lqforms

Lemma 2.1. *For any $\zeta \in \mathbb{R}^N$, we have*

$$L_H^\varphi(\sigma,\zeta) = (\sigma,\zeta)\left\{\frac{\beta}{N}\left(\sigma, \widetilde{H}\sigma - H\varphi\right) - \frac{N-1}{N}\right\} - \beta\left(\zeta, \widetilde{H}\sigma - H\varphi\right). \tag{2.10}$$

e:LS-lform

For any self-adjoint matrix C and $\zeta \in \mathbb{R}^N$, we have

$$L_H^\varphi \log\left(\frac{1}{2}(\sigma-\zeta)_C^2\right) = -2\frac{(\sigma, C(\sigma-\zeta))}{(\sigma-\zeta)_C^2}\left\{\frac{N-1}{N} - \frac{\beta}{N}(\sigma,\widetilde{H}\sigma) + \frac{\beta}{N}(\sigma, H\varphi)\right\} \tag{2.11}$$

e:LS-qform

$$+ 2\frac{\mathrm{Tr}\, C - \beta\left(\widetilde{H}\sigma - H\varphi, C(\sigma-\zeta)\right) - \frac{1}{N}(\sigma)_C^2}{(\sigma-\zeta)_C^2} - \frac{\Gamma_1^{H,\varphi}\left((\sigma-\zeta)_C^2\right)}{\left[(\sigma-\zeta)_C^2\right]^2}. \tag{2.12}$$

e:LS-qform

Proof. Write Tr_{N-1} the trace on the sphere and Tr_N that of Euclidean space. For linear forms we can write

$$L_H^\varphi(\sigma,\zeta) = \mathrm{Tr}_{N-1}\left(\nabla_S^2(\sigma,\zeta)\right) - \beta\left(\pi_\sigma\left(\widetilde{H}\sigma - H\varphi\right)\right)\cdot\left(\nabla_S(\sigma,\zeta)\right)$$

$$= \mathrm{Tr}_N\left(-\frac{1}{N}(\sigma,\zeta)\pi_\sigma\right) - \beta\left(\pi_\sigma\left(\widetilde{H}\sigma - H\varphi\right)\right)\cdot(\pi_\sigma\zeta)$$

$$= -\frac{N-1}{N}(\sigma,\zeta) - \beta\left(\zeta, \widetilde{H}\sigma - H\varphi\right) + \frac{\beta}{N}(\sigma,\zeta)\left(\sigma, \widetilde{H}\sigma - H\varphi\right),$$

which is (2.10).

For quadratic forms, we may compute

$$\begin{aligned} L_H^\varphi\left(\frac{1}{2}(\sigma-\zeta)_C^2\right) &= \mathrm{Tr}_{N-1}\left(\frac{1}{2}\nabla_S^2(\sigma-\zeta)_C^2\right) - \frac{\beta}{2}\left(\pi_\sigma\left(\widetilde{H}\sigma - H\varphi\right)\right)\cdot\left(\nabla_S(\sigma-\zeta)_C^2\right) \\ &= \mathrm{Tr}_N\left(\pi_\sigma C\pi_\sigma - \frac{1}{N}(\sigma, C(\sigma-\zeta))\pi_\sigma\right) - \beta\left(\pi_\sigma\left(\widetilde{H}\sigma - H\varphi\right)\right)\cdot(\pi_\sigma C(\sigma-\zeta)) \\ &= \mathrm{Tr}\, C - \frac{1}{N}(\sigma)_C^2 - \frac{N-1}{N}(\sigma, C(\sigma-\zeta)) \\ &\quad - \beta\left(\widetilde{H}\sigma - H\varphi, C(\sigma-\zeta)\right) + \frac{\beta}{N}(\sigma, \widetilde{H}\sigma - H\varphi)(\sigma, C(\sigma-\zeta)) \\ &= \mathrm{Tr}\, C - \beta\left(\widetilde{H}\sigma - H\varphi, C(\sigma-\zeta)\right) - \frac{1}{N}(\sigma)_C^2 \\ &\quad - (\sigma, C(\sigma-\zeta))\left\{\frac{N-1}{N} - \frac{\beta}{N}(\sigma, \widetilde{H}\sigma) + \frac{\beta}{N}(\sigma, H\varphi)\right\}. \end{aligned} \tag{2.13}$$ e:LS-qform-b

Together with the chain rule,

$$L_H^\varphi \log\left(\frac{1}{2}(\sigma-\zeta)_C^2\right) = \frac{L_H^\varphi(\sigma-\zeta)_C^2}{(\sigma-\zeta)_C^2} - \frac{\Gamma_1^{H,\varphi}\left(\frac{1}{2}(\sigma-\zeta)_C^2\right)}{\left[\frac{1}{2}(\sigma-\zeta)_C^2\right]^2},$$

we are done. □

2.1. The case of zero external field.

p:ibp-scalar-spher

Proposition 2.2. *Suppose $\beta\widetilde{H} = \beta H + z\,\mathrm{id}$ with $z = z(\beta) \in \mathbb{R}$ chosen to satisfy*

$$\mathrm{Tr}\,\frac{1}{\beta\widetilde{H}} = N. \tag{2.14}$$ a:res-H=bN

Denoting $Y = Y_{\widetilde{H}}(\sigma) = \frac{\pi_\sigma \widetilde{H}^{-1}\sigma}{(\sigma)_{\widetilde{H}^{-1}}^2}$, we have the integration by parts identity

$$\mathbb{E}_{\nu_H^\varphi}\left[\left\{1 - \frac{\beta}{N}(\sigma, \widetilde{H}\sigma) + \frac{\beta}{N}(\sigma, H\varphi)\right\}\Gamma_1^{H,\varphi}(f)\right] = 2\mathbb{E}_{\nu_H^\varphi}\left[\nabla_S^2 f(\nabla_S f, Y) - |Y|^2|\nabla_S f|^2\right]. \tag{2.15}$$ a:res-H=bN

Proof. We start by applying Lemma 2.11 with $\zeta = 0$ and $C = \widetilde{H}^{-1}$ to rewrite the term (2.9),

$$\begin{aligned} \left\{1 - \frac{\beta}{N}(\sigma, \widetilde{H}\sigma) + \frac{\beta}{N}(\sigma, H\varphi)\right\} &= \frac{1}{N} - \frac{1}{2}L_H^\varphi \log\left(\frac{1}{2}(\sigma)_{\widetilde{H}^{-1}}^2\right) \\ &\quad + \frac{\mathrm{Tr}\,\widetilde{H}^{-1} - \beta\left(\widetilde{H}\sigma - H\varphi, \widetilde{H}^{-1}\sigma\right) - \frac{1}{N}(\sigma)_{\widetilde{H}^{-1}}^2}{(\sigma)_{\widetilde{H}^{-1}}^2} - 2|Y(\sigma)|^2, \end{aligned}$$

so that we may integrate by parts,

$$\begin{aligned} \mathbb{E}_{\nu_H^\varphi}\left[\left\{1 - \frac{\beta}{N}(\sigma, \widetilde{H}\sigma) + \frac{\beta}{N}(\sigma, H\varphi)\right\}|\nabla_S f|^2\right] &= \mathbb{E}_{\nu_H^\varphi}\left[\frac{\mathrm{Tr}\,\widetilde{H}^{-1} - \beta\left(\widetilde{H}\sigma - H\varphi, \widetilde{H}^{-1}\sigma\right)}{(\sigma)_{\widetilde{H}^{-1}}^2}|\nabla_S f|^2\right] \\ &\quad + \mathbb{E}_{\nu_H^\varphi}\left[Y\cdot\nabla_S|\nabla_S f|^2 - 2|Y|^2|\nabla_S f|^2\right]. \end{aligned}$$

For the case $\varphi = 0$ the numerator in the first term simplifies to

$$\mathrm{Tr}\,\widetilde{H}^{-1} - \beta|\sigma|^2 = \mathrm{Tr}\,\widetilde{H}^{-1} - \beta N, \tag{2.16}$$

which vanishes by the hypothesis (2.14). □

The field Y is actually (uniformly) quite small, i.e. of order $O(N^{-1/2})$, as long as the $\widetilde{H} \succ \epsilon_\beta \,\mathrm{id}$ for some (possibly β-dependent, but N-independent) $\epsilon_\beta > 0$. To see this, note that after diagonalizing $\widetilde{H}^{-1}$, we can write

$$\frac{1}{N}|\pi_\sigma \widetilde{H}^{-1}\sigma|^2 = \frac{1}{N}(\sigma, \widetilde{H}^{-2}\sigma) - \frac{1}{N^2}(\sigma, \widetilde{H}^{-1}\sigma)^2 = \sum_{i=1}^N \frac{\widetilde{\sigma}_i^2}{N}\lambda_i(\widetilde{H})^{-2} - \left[\sum_{i=1}^N \frac{\widetilde{\sigma}_i^2}{N}\lambda_i(\widetilde{H})^{-1}\right]^2. \tag{2.17}$$ e:Y-repr

But this is just the variance[1] of the empirical measure $\sum_{i=1}^N \frac{\widetilde{\sigma}_i^2}{N}\delta_{\lambda_i(\widetilde{H})^{-1}}$. Then, by a crude variance bound (e.g. the variance of a Ber(1/2) supported at $\{\lambda_{\max}^{-1}, \lambda_{\min}^{-1}\}$), we have

$$|Y|^2 \le \frac{\left[\frac{1}{\lambda_{\min}(\widetilde{H})} - \frac{1}{\lambda_{\max}(\widetilde{H})}\right]^2}{4N}\lambda_{\max}(\widetilde{H})^2 =: \delta^2 = O_\beta(N^{-1}), \tag{2.18}$$ e:Y-bound

for all $\sigma \in \mathbb{S}^{N-1}_{\sqrt{N}}$. Thus Proposition 2.2 leads to the following estimate on the (integrated) $\Gamma_2^{H,\varphi}$ curvature.

p:ibp-Ga2-est

Proposition 2.3. *Suppose $\beta\widetilde{H} = \beta H + z\,\mathrm{id}$ with $z = z(\beta) \in \mathbb{R}$ chosen to satisfy*

$$\mathrm{Tr}\,\frac{1}{\beta\widetilde{H}} = N. \tag{2.19}$$ e:res-H=bN-2

Further suppose that $\frac{1}{\epsilon_\beta}\,\mathrm{id} \succ \widetilde{H} \succ \epsilon_\beta\,\mathrm{id}$ for some $\epsilon_\beta > 0$ not depending on N. Then for $\nu_H = \nu_H^{\varphi=0}$ we have the lower bound,

$$\mathbb{E}_{\nu_H}\left[\Gamma_2^H(f)\right] \ge \left\{\beta\lambda_{\min}(\widetilde{H}) - O_\beta(N^{-1})\right\}\mathbb{E}_{\nu_H}\left[\Gamma_1^H(f)\right]. \tag{2.20}$$ e:res-H=bN-2

Proof. From Proposition (2.2) and the bound (2.18) we may estimate

$$\begin{aligned}
&\mathbb{E}_{\nu_H^\varphi}\left[\Gamma_2^{H,\varphi}(f)\right] \\
&= \mathbb{E}_{\nu_H^\varphi}\left[|\nabla_S^2 f|^2\right] + 2\mathbb{E}_{\nu_H^\varphi}\left[\nabla_S^2 f(\nabla_S f, Y) - |Y|^2|\nabla_S f|^2\right] - \frac{2}{N}\mathbb{E}_{\nu_H^\varphi}\left[\Gamma_1^{H,\varphi}(f)\right] + \beta\mathbb{E}_{\nu_H^\varphi}\left[|\nabla_S f|^2_{\widetilde{H}}\right] \\
&\ge \mathbb{E}_{\nu_H^\varphi}\left[|\nabla_S^2 f|^2\right] - 2\delta\sqrt{\mathbb{E}_{\nu_H^\varphi}\left[|\nabla_S^2 f|^2\right]\mathbb{E}_{\nu_H^\varphi}\left[\Gamma_1^{H,\varphi}(f)\right]} + \left\{\beta\lambda_{\min}(\widetilde{H}) - \frac{2}{N} - 2\delta^2\right\}\mathbb{E}_{\nu_H^\varphi}\left[\Gamma_1^{H,\varphi}(f)\right] \\
&\ge \left\{\beta\lambda_{\min}(\widetilde{H}) - \frac{2}{N} - 3\delta^2\right\}\mathbb{E}_{\nu_H^\varphi}\left[\Gamma_1^{H,\varphi}(f)\right],
\end{aligned}$$

where in the last line we used the fact that $a^2 - 2\delta ab \ge -\delta^2 b^2$. Since $\delta^2 = O(N^{-1})$, we are done. □

Theorem 2.4 (Poincaré inequality for SSK with no external field, $\beta < 1$)**.** *Let $0 < \beta < 1$. For the SSK measure $\nu_H = \nu_H^{\varphi=0}$ with H a standard GOE, there holds a Poincaré inequality,*

$$\mathrm{Var}_{\nu_H}(f) \le \frac{1}{\beta(1-\beta)^2 - O_\beta(N^{-0.1})}\mathbb{E}_{\nu_H}\left[|\nabla_S f|^2\right], \tag{2.21}$$ e:PI-SSK-0h

with overwhelming probability.

Proof. Proposition 2.3 left us the task of finding $z(\beta)$ for which (2.19) holds. If H is a GOE, then by rigidity

[1] Actually, we may estimate even more crudely $\|\pi_\sigma\|_{\mathrm{op}} \le 1$. This only loses a constant factor assuming $\lambda_{\min}(\widetilde{H})^{-1} - \lambda_{\max}(\widetilde{H})^{-1} \asymp \lambda_{\min}(\widetilde{H})^{-1} \asymp 1$.

of eigenvalues one has

$$\operatorname{Tr} \frac{1}{\beta H + z \operatorname{id}} = \sum_{k=1}^{N} \frac{1}{\beta \lambda_k(H) + z} \tag{2.22}$$

$$= \frac{N}{\beta} \left\{ \int \frac{d\rho_{\mathrm{sc}}(x)}{x + z/\beta} + O(N^{-1}) \right\} \tag{2.23}$$

$$= \frac{N}{\beta} \left\{ \frac{\frac{z}{\beta} - \sqrt{\left(\frac{z}{\beta}\right)^2 - 4}}{2} + O(N^{-1}) \right\} \tag{2.24}$$

$$= N \left\{ \frac{z - \sqrt{z^2 - 4\beta^2}}{2\beta^2} + O(N^{-1}) \right\}. \tag{2.25}$$

We may ignore the $O(N^{-1})$ error, and solve for $z(\beta) \in \mathbb{R}$:

$$\frac{z - \sqrt{z^2 - 4\beta^2}}{2\beta^2} = 1 \implies z = 1 + \beta^2. \tag{2.26}$$

Thus choosing $z(\beta) = 1 + \beta^2$ ensures (2.19) holds. Furthermore, from classical edge estimates for the GOE, we have

$$\beta \widetilde{H} = \beta H + (1 + \beta^2) \operatorname{id} \geq \left\{ \left(-2 - O(N^{-0.1})\right) \beta + \left(1 + \beta^2\right) \right\} \operatorname{id} \geq \left\{ (1 - \beta)^2 - O_\beta(N^{-0.1}) \right\} \operatorname{id}, \tag{2.27}$$

with overwhelming probability; this forces the threshold $\beta < 1$; the upper bound $\widetilde{H} \prec \frac{1}{\epsilon_\beta} \operatorname{id}$ is trivial and only qualitative. Hence for all $\beta < 1$, and all sufficiently large N, it holds

$$\mathbb{E}_{\nu_H} \left[\Gamma_2^H(f) \right] \geq \left\{ \beta(1 - \beta)^2 - O_\beta(N^{-0.1}) \right\} \mathbb{E}_{\nu_H} \left[\Gamma_1^H(f) \right], \tag{2.28}$$

from which the Poincaré inequality (2.21) follows. □

2.2. The case of nonzero external field. The ideas in the previous section can be adapted to the more general case of nonzero external fields, but as the proof is strictly more technical with no new conceptual additions, I leave it out to keep this document short. If you really want to see it I can send you my writeup.

3. Sherrington-Kirkpatrick model

In this section, we push the PI threshold $\beta < 1/4$ (obtained in [3]) to $\beta < 0.37$, for the SK measure,

$$\nu_{\beta,H}(d\sigma) \propto \frac{1}{Z_{\beta,H}} e^{-\frac{\beta}{2}(\sigma,H\sigma)} \prod_{x\in\Lambda} \rho_1(d\sigma_x), \tag{3.1}$$

where $H = [\![J]\!]$ is a GOE with spectrum in $[-2-\epsilon, 2+\epsilon]$, up to scalar shifts. Our approach builds off of the one-step renormalization in [3]; in particular, we provide a better estimate on the Poincaré constant $C_{\mathrm{PI}}(\nu_s)$ of the renormalized measure that leverages more than just the bare Hessian estimate,

$$\nabla^2 U_s(\sigma) = C_{s,\infty}^{-1} + \frac{1}{s}\,\mathrm{id} - \frac{1}{s^2} A_s^\sigma \succeq \left(\frac{1}{s} - \frac{1}{s^2}\right)\mathrm{id}\,. \tag{3.2}$$

In fact, we may estimate in expectation the entire (Langevin[2]) Γ_2 form,

$$\mathbb{E}_{\nu_s}\left[\Gamma_2(f)\right] = \mathbb{E}_{\nu_s}\left[|\nabla^2 f|^2 + \left(\nabla f, \left[\nabla^2 U_s\right]\nabla f\right)\right], \tag{3.3}$$ e:SK-Ga2

with respect to the Γ_1 form,

$$\mathbb{E}_{\nu_s}\left[\Gamma_1(f)\right] = \mathbb{E}_{\nu_s}\left[|\nabla f|^2\right]. \tag{3.4}$$ e:SK-Ga2

More generally, we will consider a broader class of "smoothed SK" measures with external field, of the form

$$\nu^{G,h}(d\sigma) = \nu_s^{G,h}(d\sigma) \propto \exp\left(-V_s(\sigma) - \frac{1}{2}(\sigma, G\sigma) - (\sigma, h)\right) d\sigma, \qquad \sigma \in \mathbb{R}^N, \tag{3.5}$$

where G will typically be a resolvent-type function of H, and $h \in \mathbb{R}^N$ an external field which may depend on the randomness H in some way.

In general we would like to decompose (3.3) into two (or more) blocks. For a partition $[\![N]\!] = I \sqcup J$, since $\nabla^2 V_s$ is diagonal we may write

$$\begin{aligned}
\mathbb{E}_{\nu_s}\left[\Gamma_2(f)\right] &= \mathbb{E}_{\nu_s}\left[|\nabla_I\nabla_J f|^2 + 2\left(\nabla_I f, G_{IJ}\nabla_J f\right)\right] && (3.6)\\
&\quad + \mathbb{E}_{\nu_s}\left[|\nabla_I^2 f|^2 + \left(\nabla f, \left[G_{II} + \nabla_I^2 V_s\right]\nabla f\right)\right] && (3.7)\\
&\quad + \mathbb{E}_{\nu_s}\left[|\nabla_J^2 f|^2 + \left(\nabla f, \left[G_{JJ} + \nabla_J^2 V_s\right]\nabla f\right)\right] && (3.8)\\
&= \mathbb{E}_{\nu_s}\left[|\nabla_I\nabla_J f|^2 + 2\left(\nabla_I f, G_{IJ}\nabla_J f\right)\right] && (3.9)\\
&\quad + \mathbb{E}_{\nu_J}\left[\mathbb{E}_{\mu_I^J}\left[\Gamma_{2,I}^J(f)\right]\right] && (3.10)\\
&\quad + \mathbb{E}_{\nu_I}\left[\mathbb{E}_{\mu_J^I}\left[\Gamma_{2,J}^I(f)\right]\right] && (3.11)
\end{aligned}$$

where $\nu_I = \nu_{s,I}$ denotes the I-marginal,

$$d\nu_I(\sigma_I) \propto \int_{\mathbb{R}^{|J|}} \nu_s(\sigma_I, d\sigma_J), \tag{3.12}$$

and $\mu_J^I = \mu_{s,J}^I$ denotes the conditional measure of the sites at J given those at I,

$$d\mu_J^I(\sigma_J) \propto \nu_s(\sigma_I, d\sigma_J) \propto \exp\left(-V_s(\sigma_I,\sigma_J) - \frac{1}{2}(\sigma_J, G_{JJ}\sigma_J) - (\sigma_J, G_{JI}\sigma_I)\right) d\sigma_J. \tag{3.13}$$

Then, above we denoted $\Gamma_{2,J}^I$ the iterated carré du champ operator associated to ν_J^I (and conversely). Suppose then that we are able to show, for some (positive) $M_I \in \mathbb{R}^{|I|}\times\mathbb{R}^{|I|}$ and $M_J \in \mathbb{R}^{|J|}\times\mathbb{R}^{|J|}$,

$$\mathbb{E}_{\mu_I^J}\left[\Gamma_{2,I}^J(f)\right] \geq \mathbb{E}_{\mu_I^J}\left[\left(\nabla_I f, M_I\nabla_I f\right)\right], \tag{3.14}$$

$$\mathbb{E}_{\mu_J^I}\left[\Gamma_{2,J}^I(f)\right] \geq \mathbb{E}_{\mu_J^I}\left[\left(\nabla_J f, M_J\nabla_J f\right)\right]. \tag{3.15}$$

[2]Though Γ_2 here depends on s, we choose to omit the dependence as s is fixed and ν_s is treated as the equilibrium measure. Also this prevents confusing with the dynamical $\Gamma_{k,s}$ operators associated to the non-equilbrium Polchinski dynamics.

Then we could insert back into the full Γ_2 decomposition to get

$$\mathbb{E}_{\nu_s}\left[\Gamma_2(f)\right] \geq \mathbb{E}_{\nu_s}\left[|\nabla_I \nabla_J f|^2\right] + \mathbb{E}_{\nu_s}\left[\left(\begin{bmatrix}\nabla_I f \\ \nabla_J f\end{bmatrix}, \begin{bmatrix}M_I & G_{IJ} \\ G_{JI} & M_J\end{bmatrix}\begin{bmatrix}\nabla_I f \\ \nabla_J f\end{bmatrix}\right)\right].$$

My original intuition for approaching the hypercube SK model was that, each half-block system is effectively at temperature $\beta/\sqrt{2}$, and so we should be able to bootstrap bounds on the (conditional, child) $\Gamma_{2,I}^J$ at higher temperatures, to bounds on the (full, parent) Γ_2 at lower temperatures. Instead, for the $\beta < 0.37$ proof we do a (probably) crude decomposition into N single-site blocks (compare with Wang's two-site analysis [18]); this only uses the diagonal terms G_{ii} in the local law; perhaps we need to leverage the full isotropic/entrywise local law (so off-diagonals G_{ij}) to do better). For the $\beta < 0.37$ proof, below one may take $I = [\![N]\!]$ and $J = \varnothing$ in Lemma 3.2 and throughout.

l:Ga2-lyapunov

Lemma 3.1 (Second order Lyapunov function estimate)**.** *Let $\nu(d\sigma) \propto e^{-U(\sigma)}d\sigma$ be a probability measure on $\mathbb{R}^n$ and let*

$$L = \Delta - \nabla U \cdot \nabla \tag{3.16}$$

denote the generator of the (reversible) Langevin dynamics associated to ν, and Γ_2 the iterated carré du champ operator associated to L. For any smooth functions $w_i : \mathbb{R}^n \to \mathbb{R}$, $1 \leq i \leq n$, we have the lower bound,

$$\mathbb{E}_\nu\left[\Gamma_2(f)\right] \geq \mathbb{E}_\nu\left[\left(\nabla f, \left[\nabla^2 U - \operatorname*{diag}_{1\leq i\leq n}\left(\frac{Lw_i}{w_i}\right)\right]\nabla f\right)\right]. \tag{3.17}$$

e:Ga2-lyapunov

Proof. Integrating by parts in each coordinate $1 \leq i \leq n$, we have

$$-\mathbb{E}_\nu\left[(\partial_i f)^2\frac{Lw_i}{w_i}\right] = 2\mathbb{E}_\nu\left[\frac{\partial_i f}{w_i}\nabla\partial_i f\cdot\nabla w_i\right] - \mathbb{E}_\nu\left[\frac{(\partial_i f)^2}{w_i^2}|\nabla w_i|^2\right].$$

Thus, we may complete the square against the Hessian square term in $\Gamma_2(f)$, which yields

$$\begin{aligned}\mathbb{E}_\nu\left[|\nabla^2 f|^2\right] + \mathbb{E}_\nu\left[\left(\nabla f, \operatorname*{diag}_{1\leq i\leq n}\left(\frac{Lw_i}{w_i}\right)\nabla f\right)\right] &= \sum_{i=1}^n \mathbb{E}_\nu\left[|\nabla\partial_i f|^2 + (\partial_i f)^2\frac{Lw_i}{w_i}\right] \\ &= \mathbb{E}_\nu\left[\left|\nabla\partial_i f - \frac{\partial_i f}{w_i}\nabla w_i\right|^2\right] \geq 0,\end{aligned}$$

from which (3.17) follows. □

l:Ga2IJ-lyapunov

Lemma 3.2. *Let $I \sqcup J = [\![N]\!]$ be any partition and retain the notation from above. Let $0 \leq \eta \leq 1$ and set*

$$p_*(\eta) = 2\eta^2 - \eta + 2(1-\eta)\sqrt{\eta(1+\eta)}. \tag{3.18}$$

e:Ga2-lyapunov

Then we have the lower bound

$$\mathbb{E}_{\mu_I^J}\left[\Gamma_{2,I}^J(f)\right] \geq \mathbb{E}_{\mu_I^J}\left[\left(\nabla_I f, \left[G_{II} + \operatorname*{diag}_{i\in I}\left(\frac{1-\eta}{s} - \eta G_{ii} + \frac{p_*(\eta)-1}{s^2}\right)\right]\nabla_I f\right)\right]. \tag{3.19}$$

e:Ga2IJ-lyapunov

Proof. We decompose μ_I^J further into one-site conditional measures. For $i \in I$ denote $(i) = \{i\}^c$ and note that

$$\mu_i(d\sigma_i) := \mu_i^{(i)}(d\sigma_i) \propto \underbrace{\exp\left(-v_s(\sigma_i) - \frac{1}{2}G_{ii}\sigma_i^2 - \left(\sigma_i, G_{i,(i)}\sigma_{(i)}\right)\right)}_{=:e^{-u_s(\sigma_i)}} d\sigma_i \tag{3.20}$$

e:Ga2IJ-lyapunov

satisfies $\mathbb{E}_{\mu_I^J} = \mathbb{E}_{\nu_{(i)}^J}\mathbb{E}_{\mu_i^{(i)}}$, where $\nu_{(i)}^J$ is the (i)-marginal of μ_I^J. Let $\Gamma_{2,i}$ denote the (one-dimensional) iterated carré du champ operator associated to μ_i, i.e.

$$\Gamma_{2,i}(f) = \left(\partial_{\sigma_i}^2 f\right)^2 + \left[\partial_{\sigma_i}^2 v_s + G_{ii}\right]\cdot\left(\partial_{\sigma_i} f\right)^2. \tag{3.21}$$

We may break $\Gamma_{2,I}^J$ apart into $\Gamma_{2,i}$,

$$\Gamma_{2,I}^J(f) = |\nabla_I^2 f|^2 + \left(\nabla_I f, \left[\nabla_I^2 V_s + G_{II}\right] \nabla_I f\right) \tag{3.22}$$

$$= (\nabla_I f, G_{II} \nabla_I f) + \sum_{i \in I} \left[\Gamma_{2,i}(f) - (\partial_i f)^2 G_{ii}\right] + \sum_{\substack{i_1, i_2 \in I \\ i_1 \neq i_2}} (\partial_{i_1 i_2} f)^2 . \tag{3.23}$$ e:Ga2IJ-Ga2i

So, aside from the positive G_{II} term, an estimate on $\mathbb{E}_{\mu_I^J} \Gamma_{2,I}^J(f)$ boils down to an estimate on

$$\begin{aligned}\mathbb{E}_{\mu_i}\left[\Gamma_{2,i}(f) - (\partial_i f)^2 G_{ii}\right] &= \mathbb{E}_{\mu_i}\left[\left(\partial_i^2 f\right)^2 + \left(\partial_i^2 v_s\right) \cdot \left(\partial_i f\right)^2\right] \\ &= \mathbb{E}_{\mu_i}\left[(\partial_i^2 f)^2 + \left(\frac{1}{s} - \frac{1}{s^2}\operatorname{sech}^2(\sigma_i/s)\right) \cdot (\partial_i f)^2\right].\end{aligned}$$

Denoting ℓ_i the Langevin generator associated to μ_i,

$$\begin{aligned}\ell_i &= \partial_i^2 - \left(\partial_i v_s + G_{ii}\sigma_i + \sum_{j \neq i} G_{ij}\sigma_j\right)\partial_i \\ &= \partial_i^2 - \left(\frac{1}{s}\left(\sigma_i - \tanh\left(\sigma_i/s\right)\right) + G_{ii}\sigma_i + \sum_{j \neq i} G_{ij}\sigma_j\right)\partial_i,\end{aligned}$$

we will supply a Lyapunov function $w : \mathbb{R} \to \mathbb{R}$

$$\partial_i^2 u_s - \frac{\ell_i w}{w} \geq (1-\eta)\left(G_{ii} + \frac{1}{s}\right) + \frac{p_*(\eta) - 1}{s^2}, \tag{3.24}$$ c:lyapunov-ell

where $p_*(\eta)$ is defined below. In particular, we consider the following family of functions, for $0 \leq \eta \leq 1$ and $p \in \mathbb{R}$,

$$w_{i,\eta,p}(\sigma) = \exp\left(\eta\left\{\left(G_{ii} + \frac{1}{s}\right)\frac{\sigma_i^2}{2} + \left(\sigma_i, G_{i,(i)}\sigma_{(i)}\right)\right\}\right)\operatorname{sech}^p(\sigma_i/s). \tag{3.25}$$ c:lyapunov-ell

We compute:

$$\begin{aligned}\partial_i w_{i,\eta,p} &= \eta\left\{\sigma_i\left(G_{ii} + \frac{1}{s}\right) + G_{i,(i)}\sigma_{(i)}\right\} w_{i,\eta,p} - \frac{p}{s}\tanh(\sigma_i/s) w_{i,\eta,p}, \\ \partial_i^2 w_{i,h,p} &= \eta\left(G_{ii} + \frac{1}{s}\right) w_{i,\eta,p} + \eta^2\left\{\sigma_i\left(G_{ii} + \frac{1}{s}\right) + G_{i,(i)}\sigma_{(i)}\right\}^2 w_{i,\eta,p} \\ &\quad - 2 \cdot \frac{p}{s}\tanh(\sigma_i/s) \cdot \eta\left\{\sigma_i\left(G_{ii} + \frac{1}{s}\right) + G_{i,(i)}\sigma_{(i)}\right\} w_{i,\eta,p} \\ &\quad - \frac{p}{s^2}\operatorname{sech}^2(\sigma_i/s) w_{i,\eta,p} + \frac{p^2}{s^2}\tanh^2(\sigma_i/s) w_{i,\eta,p}.\end{aligned}$$

Thus, denoting $Q_i = \sigma_i(G_{ii} + \frac{1}{s}) + G_{i,(i)}\sigma_{(i)}$ and $\tau_i = \tanh(\sigma_i/s)$, we have

$$\begin{aligned}\frac{\ell_i w_{i,\eta,p}}{w_{i,\eta,p}} &= \eta\left(G_{ii} + \frac{1}{s}\right) + \eta^2 Q_i^2 \\ &\quad - \frac{p}{s^2}\operatorname{sech}^2(\sigma_i/s) - 2\eta\frac{p}{s}\tanh(\sigma_i/s) Q_i + \frac{p^2}{s^2}\tau_i^2 \\ &\quad - \left(Q_i - \frac{1}{s}\tanh(\sigma_i/s)\right)\left(\eta Q_i - \frac{p}{s}\tanh(\sigma_i/s)\right) \\ &= -\eta(1-\eta)Q_i^2 + \frac{\tau_i}{s}\left(\eta + p - 2\eta p\right) Q_i + \eta\left(G_{ii} + \frac{1}{s}\right) - \frac{p}{s^2} + \frac{p^2}{s^2}\tau_i^2.\end{aligned}$$

Now, minimizing over $Q_i \in \mathbb{R}$, we have

$$\begin{aligned}\left(\partial_i^2 v_s\right) + G_{ii} - \frac{\ell_i w_{i,\eta,p}}{w_{i,\eta,p}} &= \eta(1-\eta)Q_i^2 + \frac{\tau_i}{s}(2\eta p - \eta - p)Q_i + (1-\eta)\left(G_{ii} + \frac{1}{s}\right) + \frac{p}{s^2} - \frac{p^2}{s^2}\tau_i^2 - \frac{1-\tau_i^2}{s^2} \\ &\geq (1-\eta)\left(G_{ii} + \frac{1}{s}\right) + \frac{p}{s^2} - \frac{p^2}{s^2}\tau_i^2 - \frac{1-\tau_i^2}{s^2} - \frac{\tau_i^2}{s^2}\frac{(2\eta p - \eta - p)^2}{4\eta(1-\eta)} \\ &= (1-\eta)\left(G_{ii} + \frac{1}{s}\right) + \frac{p-1}{s^2} + \frac{\tau_i^2}{s^2}\left(1 - p^2 - \frac{(2\eta p - \eta - p)^2}{4\eta(1-\eta)}\right).\end{aligned}$$

We will choose $p_* = p_*(\eta)$ to vanish the coefficient of the τ_i^2 term, i.e.

$$1 - p_*^2 = \frac{(2\eta p_* - \eta - p_*)^2}{4\eta(1-\eta)} \implies p_*(\eta) = 2\eta^2 - \eta + 2(1-\eta)\sqrt{\eta(1+\eta)}. \tag{3.26}$$

With this choice of p, we have (uniformly in $\sigma \in \mathbb{R}^N$)

$$\partial_i^2 u_s - \frac{\ell_i w_{i,\eta,p_*}}{w_{i,\eta,p_*}} \geq (1-\eta)\left(G_{ii} + \frac{1}{s}\right) + \frac{p_*(\eta) - 1}{s^2}, \tag{3.27}$$

and so by Lemma 3.1, we have

$$\mathbb{E}_{\mu_i}\left[\Gamma_{2,i}(f) - (\partial_i f)^2 G_{ii}\right] \geq \mathbb{E}_{\mu_i}\left[(\partial_i f)^2\left(\frac{1-\eta}{s} - \eta G_{ii} + \frac{p_*(\eta) - 1}{s^2}\right)\right]. \tag{3.28}$$

Inserting this estimate back into the decomposition (3.23) yields (3.19) □

Now, before we optimize in the parameter $0 \leq \eta \leq 1$, we need a good estimate on the GOE resolvents which appear in (3.19). For the case

$$G = C_{s,\infty}^{-1} = \frac{1}{\frac{1}{\beta H} - s\,\mathrm{id}} = \frac{\beta H}{\mathrm{id} - s\beta H}, \tag{3.29}$$ c:G=Csinfty-1

the following lemma provides bounds on G_{II} and G_{ii}. Note that we always assume that $\beta H = \beta J + c\,\mathrm{id} > 0$ and $C_{s,\infty}^{-1} > 0$ which translates to the constraints

$$-2\beta + c > 0, \qquad \frac{1}{s} > 2\beta + c. \tag{3.30}$$ c:pol-cov-pos

In particular, this ensures that the parameter $E_{c,s,\beta}$ as defined below is always negative.

Lemma 3.3 (GOE resolvent estimate)**.** *Let $G = C_{s,\infty}^{-1}$ as in (3.29), and fix a (large) $D > 0$ and (small) $\epsilon > 0$. Define the location parameter,*

$$E_{c,s,\beta} := \frac{c - 1/s}{\beta} < 0, \tag{3.31}$$ c:pol-cov-pos

and suppose that it satisfies

$$\mathrm{dist}\left(E_{c,s,\beta}, [-2,2]\right) > \epsilon_{c,s,\beta}, \tag{3.32}$$ c:Ecsb-out

for some $\epsilon_{c,s,\beta} > 0$ that is independent of N. Then for all sufficiently large N, it holds

$$\mathbb{P}_J\left(\max_{1\leq i\leq N}\left|\left[C_{s,\infty}^{-1}\right]_{ii} - \left(-\frac{1}{s} - \frac{1}{s^2\beta}\frac{E_{c,s,\beta} + \sqrt{E_{c,s,\beta}^2 - 4}}{2}\right)\right| \leq O_{c,s,\beta}\left(\frac{N^\epsilon}{\sqrt{N}}\right)\right) \geq 1 - N^{-D}. \tag{3.33}$$ e:Gii-LL-est

Proof. The (entrywise, outside the spectrum) local law (e.g. [7, Theorem 10.3] suffices for our purposes. In particular, recall that it states, for any (large) $D > 0$ and all sufficiently large N,

$$\mathbb{P}\left(\max_{1\leq i\leq N}\left|\left[\frac{1}{J - zI}\right]_{ii} - \frac{1}{N}\,\mathrm{Tr}\,\frac{1}{J - zI}\right| \leq \frac{N^\epsilon}{\sqrt{N}(\kappa+\eta)^{1/4} + (\kappa+\eta)^2}\right) > 1 - N^{-D} \tag{3.34}$$ e:LL-out

where we denote $z = E + \mathrm{i}\eta$ and $\kappa = \mathrm{dist}(E, [-2,2])$ is the distance from the spectrum. In particular, for us we may take $\kappa > \epsilon_\beta > 0$ a fixed (β-dependent, but N independent) positive distance, so the bound is stable

as $\eta \downarrow 0$. Now, for our particular resolvent $G = C_{s,\infty}^{-1}$ depending on the shifted GOE matrix $\beta H = \beta J + c\,\mathrm{id}$, we write

$$\begin{aligned} C_{s,\infty}^{-1} = \frac{\beta H}{\mathrm{id} - s\beta H} &= -\frac{1}{s}\,\mathrm{id} + \frac{1}{s}\frac{1}{\mathrm{id} - s\beta H} \\ &= -\frac{1}{s}\,\mathrm{id} + \frac{1}{s^2\beta}\frac{1}{\left(\frac{1}{s\beta} - \frac{c}{\beta}\right)\mathrm{id} - J}. \end{aligned}$$

Then the local law (3.34) tells us that,

$$\mathbb{P}\left(\max_{1\le i\le N}\left|\left[C_{s,\infty}^{-1}\right]_{ii} - \frac{1}{N}\operatorname{Tr} C_{s,\infty}^{-1}\right| \le \frac{1}{s^2\beta}\frac{N^{\epsilon}}{\sqrt{N}\sqrt[4]{\epsilon_{c,s,\beta}}}\right) \ge 1 - N^{-D}, \tag{3.35}$$ e:LL-out

provided that (3.32) holds. Now, by rigidity of eigenvalues, we may estimate $\frac{1}{N}\operatorname{Tr} C_{s,\infty}^{-1}$ explicitly as well. In particular, with overwhelming probability it holds

$$\begin{aligned} \frac{1}{N}\operatorname{Tr} C_{s,\infty}^{-1} &= \int_{-2}^{2}\frac{\beta x + c}{1 - s(\beta x + c)}d\rho_{\mathrm{sc}}(x) + O\left(\frac{N^{\epsilon}}{N}\right) \\ &= -\frac{1}{s} + \frac{1}{s}\int_{-2}^{2}\frac{1}{1 - s(\beta x + c)}d\rho_{\mathrm{sc}}(x) + O\left(\frac{N^{\epsilon}}{N}\right) \\ &= -\frac{1}{s} + \frac{1}{s^2\beta}\int_{-2}^{2}\frac{1}{\left(\frac{1}{s\beta} - \frac{c}{\beta}\right) - x}d\rho_{\mathrm{sc}}(x) + O\left(\frac{N^{\epsilon}}{N}\right) \\ &= -\frac{1}{s} - \frac{1}{s^2\beta}\frac{E_{c,s,\beta} + \sqrt{E_{c,s,\beta}^2 - 4}}{2} + O\left(\frac{N^{\epsilon}}{N}\right). \end{aligned}$$

Altogether, the estimate (3.33) follows. □

Theorem 3.4. *Fix $D > 0$. For every $\beta < 0.37$, there exists a constant C_β depending only on β (in particular, not on N), such that for all sufficiently large (depending on D only) N it holds*

$$\mathbb{P}_J\left(C_{PI}(\nu_{\beta,H}) < C_\beta\right) > 1 - N^{-D} \tag{3.36}$$

Proof. We aim to apply Lemma 3.2 with the case $I = [\![N]\!]$ and $J = \varnothing$. Let us study the diagonal curvature term,

$$\Phi_{s,i}(\eta) := \frac{1-\eta}{s} - \eta G_{ii} + \frac{p_*(\eta) - 1}{s^2} \tag{3.37}$$

over $0 \le \eta \le 1$. Since this is 1-Lipschitz in G_{ii}, the local law estimate (3.33) allows us replace G_{ii} with $\frac{1}{N}\operatorname{Tr} G$ at the loss of an arbitrarily small margin $\delta_N\,\mathrm{id}$ that vanishes as $N \to \infty$. That is,

$$\operatorname*{diag}_{1\le i\le N}\Phi_{s,i}(\eta) \succeq (\Phi_s(\eta) - \delta_N)\,\mathrm{id}, \tag{3.38}$$

where

$$\Phi_s(\eta) := \frac{1-\eta}{s} - \frac{\eta}{N}\operatorname{Tr} G + \frac{p_*(\eta) - 1}{s^2}. \tag{3.39}$$

For the rest of the proof, we make this substitution freely and note that the value (and sign) of δ_N may vary from line to line, but it always vanishing in N. Differentiating, we have

$$\Phi_s'(\eta) = -\frac{1}{s} - \frac{1}{N}\operatorname{Tr} G + \frac{1}{s^2}p_*'(\eta), \tag{3.40}$$

where the function,

$$p_*'(\eta) = 4\eta - 1 + \frac{1 - \eta - 4\eta^2}{\sqrt{\eta(1+\eta)}}, \tag{3.41}$$

is positive and strictly decreasing on $\eta \in (0, 1]$, with

$$p_*''(\eta) = 4 - \frac{1 + 3\eta + 12\eta^2 + 8\eta^3}{2\left(\eta(1+\eta)\right)^{3/2}} < 0, \qquad 0 < \eta < 1. \tag{3.42}$$

Hence the maximum in $(0,1]$ occurs at $\eta_* = \eta_*(s,G)$ given by

$$\eta_* = \begin{cases} 1 & \text{if } s + \frac{s^2}{N}\operatorname{Tr} G \le p_*'(1) = 3 - 2\sqrt{2}, \\ \text{the unique } 0 < \eta < 1 \text{ s.t. } s + \frac{s^2}{N}\operatorname{Tr} G = p_*'(\eta) & \text{o.w.} \end{cases} \tag{3.43}$$

Now, if $\eta_* = 1$ then $\Phi_s(\eta_*) = -\frac{1}{N}\operatorname{Tr} G$ which is undesirable since we require $G = C_{s,\infty}^{-1} > 0$. So we must have $s + \frac{s^2}{N}\operatorname{Tr} G > 3 - 2\sqrt{2}$, which by (3.33) translates to the condition

$$s + s^2\left(-\frac{1}{s} - \frac{1}{s^2\beta}\frac{E_{c,s,\beta} + \sqrt{E_{c,s,\beta}^2 - 4}}{2}\right) > 3 - 2\sqrt{2}. \tag{3.44}$$ c:R>3-2sqrt2

When this holds, we may express $\frac{1}{N}\operatorname{Tr} G$ in terms of the maximizer η_*, which after the approximation (3.33) says

$$\left(-\frac{1}{s} + \frac{1}{s^2\beta}\frac{E_{c,s,\beta} + \sqrt{E_{c,s,\beta}^2 - 4}}{2}\right) = \frac{p_*'(\eta_*) - s}{s^2} + \delta_N. \tag{3.45}$$ c:R>3-2sqrt2

As $N \to \infty$ and solving for c, this suggests that we take $c = c_*(s,\beta,\eta)$ where

$$E_{c_*,s,\beta} = \frac{1 - \beta^2 p_*'(\eta)^2}{\beta p_*'(\eta)}, \qquad \text{i.e.} \qquad c_*(s,\beta,\eta) := \frac{1}{s} - \frac{1 + \beta^2 p_*'(\eta)^2}{p_*'(\eta)}. \tag{3.46}$$ e:c*

In particular, this ensures that the condition (3.44) is automatically satisfied provided we end up picking an interior $\eta \in (0,1)$. With this choice of $c = c_*$, and assuming the Polchinski covariance constraints (3.30) hold, the lower edge of $G = C_{s,\infty}^{-1}$ is given by

$$\begin{aligned} \lambda_{\min}(C_{s,\infty}^{-1}) &= \lambda_{\min}\left(\frac{\beta H + c_* \operatorname{id}}{1 - s(\beta H + c_* \operatorname{id})}\right) \\ &= \frac{-2\beta + c_*}{1 - s(-2\beta + c_*)} \\ &= -\frac{1}{s} + \frac{1}{s}\frac{1}{1 - s(-2\beta + c_*)}, \\ &= -\frac{1}{s} - \frac{1}{s^2}\frac{p_*'(\eta)}{(\beta p_*'(\eta) + 1)^2}. \end{aligned}$$

To ensure (3.30) hold, we need

$$\begin{cases} \frac{1}{s} > 2\beta + \frac{1 + \beta^2 p_*'(\eta)^2}{p_*'(\eta)}, \\ 0 \ge 2\beta - \frac{1 + \beta^2 p_*'(\eta)^2}{p_*'(\eta)}. \end{cases} \tag{3.47}$$ e:c*

So let $s = s_*(\beta,\eta)$ be given by

$$s_*(\beta,\eta) := \left(2\beta + \frac{1 + \beta^2 p_*'(\eta)^2}{p_*'(\eta)}\right)^{-1} - \delta_N, \tag{3.48}$$

which we note is always positive (for sufficiently large N), ensuring the first condition. The second condition is automatic since $p_*' > 0$ on $(0,1)$ so that

$$2\beta - \frac{1 + \beta^2 p_*'(\eta)^2}{p_*'(\eta)} = -\frac{(1 - \beta p_*'(\eta))^2}{p_*'(\eta)} < 0. \tag{3.49}$$

Now, with this choice of $s = s_*(\beta, \eta)$ and $c = c_*(s_*, \beta, \eta)$, we have

$$C_{s,\infty}^{-1} + \operatorname*{diag}_{1\le i\le N} \Phi_{s,i}(\eta_i) \succeq C_{s,\infty}^{-1} + (\Phi_s(\eta) - \delta_N)\,\mathrm{id} \tag{3.50}$$

$$\succeq \left\{ \lambda_{\min}\left(C_{s,\infty}^{-1}\right) + \frac{1-\eta}{s} - \frac{\eta}{N}\operatorname{Tr} G + \frac{p_*(\eta)-1}{s^2} \right\} \mathrm{id} - \delta_N\,\mathrm{id} \tag{3.51}$$

$$\succeq \left\{ \lambda_{\min}\left(C_{s,\infty}^{-1}\right) + \frac{1-\eta}{s} - \eta\frac{p_*'(\eta)-s}{s^2} + \frac{p_*(\eta)-1}{s^2} \right\} \mathrm{id} - \delta_N\,\mathrm{id} \tag{3.52}$$

$$= \frac{1}{s^2}\left\{ p_*(\eta) - \eta p_*'(\eta) - 1 - \frac{p_*'(\eta)}{\left(\beta p_*'(\eta)+1\right)^2} \right\} \mathrm{id} - \delta_N\,\mathrm{id}\,. \tag{3.53}$$ e:Ga2-lya-eta-bou

Now, we may solve for the critical β_c by setting

$$0 = p_*(\eta) - \eta p_*'(\eta) - 1 - \frac{p_*'(\eta)}{\left(\beta p_*'(\eta)+1\right)^2}, \tag{3.54}$$ e:Ga2-lya-eta-bou

which yields

$$\beta_c = \sup_{0<\eta<1} \frac{\sqrt{\frac{p_*'(\eta)}{p_*(\eta)-\eta p_*'(\eta)-1}} - 1}{p_*'(\eta)}. \tag{3.55}$$

The numerical solution gives (approximately)

$$\beta_c = 0.370173202543, \tag{3.56}$$

achieved when $\eta = \eta_c = 0.180366655376$. Thus for any $\beta < \beta_c$ there exists an $\eta \in (0,1)$ which makes (3.53) strictly positive with a margin depending only on β. Thus, it follows from Lemma 3.2 (with $I = [\![N]\!]$ and $J = \varnothing$) that a Poincaré inequality holds for the SK measure $\nu_{\beta,H}$, with overwhelming probability in the GOE disorder. □

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